

Basic Topology

1.1 度量空间

Definition 1.1.1

令 $X \neq \emptyset$ 是一个非空集合. X 上的一个度量, 是指一个映射

$$\begin{aligned} d : X \times X &\rightarrow \mathbb{R} \\ (x, y) &\mapsto d(x, y) \end{aligned}$$

使得对于任意的 $x, y, z \in X$, 有

1. $d(x, y) \geq 0$, 且 $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) \leq d(x, y) + d(y, z)$

Example 1.1.2

$X \neq \emptyset$, 定义 $d : X \times X \rightarrow \mathbb{R}$

$$d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

是一个度量.

Proof. 1.2. 是显然的. 3. 当 $x = z$ 时是显然的. 若 $x \neq z$, 则要么 $x \neq y$, 要么 $z \neq y$, 从而

$$d(x, z) = 1 \leq d(x, y) + d(y, z)$$

□

Exercise 1.1.3

设 (X, d) 是度量空间, 定义

$$\rho : X \times X \rightarrow \mathbb{R}$$

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

则 ρ 是一个有界度量.

Proof. 1.2. 是显然的, 验证度量的第三条定义,

$$\begin{aligned} \frac{d(x, z)}{1 + d(x, z)} + \frac{d(y, z)}{1 + d(y, z)} &= \frac{2d(x, z)d(y, z) + d(x, z) + d(y, z)}{d(x, z)d(y, z) + d(x, z) + d(y, z) + 1} \\ &\geq \frac{d(x, z)d(y, z) + d(x, z) + d(y, z)}{d(x, z)d(y, z) + d(x, z) + d(y, z) + 1} \end{aligned}$$

其中最后一行小于等于 1, 由 $d(x, z) + d(y, z) \geq d(x, y)$, 利用不等式

$$\frac{a + c}{b + c} \geq \frac{a}{b}, \quad (0 < a \leq b, c \geq 0)$$

可得上式最后一行大于等于

$$\frac{d(x, z)d(y, z) + d(x, y)}{d(x, z)d(y, z) + d(x, y) + 1} \geq \frac{d(x, y)}{d(x, y) + 1}$$

□

1.2 Exercises

Problem 1.2.1 ► Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin.

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphic to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 1.2.2

X is compact, Y is Hausdorff. If $\exists f : X \rightarrow Y$ is a continuous bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$. Since K is compact, $\text{dist}(K, 0) < \infty$. If $x \neq 0$, then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \text{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction. For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx , we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality, we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left((N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \text{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continuous, only need to show it is continuous at 0. In fact, $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$. The same skills can be used to show ψ is continuous at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 1.2.3

Let X, X' be topological spaces and $x \in X, x' \in X'$ be base points. Prove that the wedge sum $X \vee X'$ is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$

Proof. $X \vee X' = (X \amalg X') / \{x \sim x'\}$. Considering map $\varphi : X \times X' \rightarrow X \vee X'$

$$\varphi(x_1, x_2) = \begin{cases} x_1, & x_2 = x' \\ x_2, & x_1 = x \end{cases}$$

□

Problem 1.2.4

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to :

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translation.

Problem 1.2.5 ▶ Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projective space $\mathbb{K}\mathbb{P}^n$ is the orbit space of $\mathbb{K}^{n+1} \setminus \{0\}$ under the action of $\mathbb{K}^*$ by rescaling.

1. Prove that $\mathbb{R}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{C}\mathbb{P}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$, where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.
3. Prove that $\mathbb{R}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{C}\mathbb{P}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{R}\mathbb{P}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \rightarrow \mathbb{R}\mathbb{P}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{R}\mathbb{P}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such that $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$. Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$. The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \rightarrow \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \rightarrow \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$, where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x(\mathbb{S}^1)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to show that $\mathbb{S}^1/\{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \rightarrow \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homeomorphism.

- (b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

□

Problem 1.2.6

The torus $\mathbb{T}$ is the quotient of $[0, 1]^2$ by the equivalence relation generated by $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$ for all $x, y \in [0, 1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof. 1. Let

$$f(x, y) := (e^{2\pi xi}, e^{2\pi yi})$$

Then

$$f(x, 0) = (e^{2\pi ix}, 1) = f(x, 1)$$

$$f(0, y) = (1, e^{2\pi yi}) = f(1, y)$$

Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x, y)]) = (e^{2\pi xi}, e^{2\pi yi})$, which is a continuous function. It is not hard to see that $\bar{f} : \mathbb{T} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1$ is a bijection. As a continuous image of $[0, 1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V . Since $\mathbb{S}^1 \times \mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.

2. Let

$$f(x, y) = (x, y) / \mathbb{Z}^2$$

Then Since $(x, 0) - (x, 1) \in \mathbb{Z}^2, (0, y) - (1, y) \in \mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1 \times \mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2 / \mathbb{Z}^2$. If $f(x_1, y_1) = f(x_2, y_2)$, then

$$(x_1 - x_2, y_1 - y_2) \in \mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\}, \text{ and } y_1 = y_2 \text{ or } \{y_1, y_2\} = \{0, 1\}$$

thus $(x_1, y_1) \sim (x_2, y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2 / \mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homomorphism. □

Problem 1.2.7

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R}) / O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R})/O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$